

# LA Shit

## Equivalence Statements

- a)  $A$  is invertible.
- b)  $A\mathbf{x} = \mathbf{0}$  has only the trivial solution.
- c) The reduced row echelon form of  $A$  is  $I_n$ .
- d)  $A$  is expressible as a product of elementary matrices.
- e)  $A\mathbf{x} = \mathbf{b}$  is consistent for every  $n \times 1$  matrix  $\mathbf{b}$ .
- f)  $A\mathbf{x} = \mathbf{b}$  has exactly one solution for every  $n \times 1$  matrix  $\mathbf{b}$ .
- g)  $\det(A) \neq 0$
- h) The column vectors of  $A$  are linearly independent.
- i) The row vectors of  $A$  are linearly independent.
- j) The column vectors of  $A$  span  $R^n$ .
- k) The row vectors of  $A$  span  $R^n$ .
- l) The column vectors of  $A$  form a basis for  $R^n$ .
- m) The row vectors of  $A$  form a basis for  $R^n$ .
- n)  $A$  has rank  $n$ .
- o)  $A$  has nullity 0.
- p) The orthogonal complement of the null space of  $A$  is  $R^n$ .
- q) The orthogonal complement of the row space of  $A$  is  $\{\mathbf{0}\}$ .
- r)  $\lambda = 0$  is not an eigenvalue of  $A$ .
- s)  $A^T A$  is invertible.
- t) The kernel of  $T_A$  is  $\{\mathbf{0}\}$ .
- u) The range of  $T_A$  is  $R^n$ .
- v)  $T_A$  is one-to-one.

## Chapter 1

### Elementary

- a)  $A^{-1} = E_k \dots E_2 E_1 I_n$
- b)  $A = E_1^{-1} E_2^{-1} \dots E_k^{-1}$

### Powers

- a)  $A^0 = I$
- b)  $A^n = AA \dots A$
- c)  $A^{-n} = (A^n)^{-1} = (A^{-1})^n = A^{-1} A^{-1} \dots A^{-1}$
- d)  $A^r A^s = A^{r+s}$

e)  $(kA)^{-1} = k^{-1}A^{-1}$

## Transpose

- a)  $(A^T)^T = A$
- b)  $(A + B)^T = A^T + B^T$
- c)  $(kA)^T = kA^T$
- d)  $(AB)^T = B^T A^T$
- e)  $(A^T)^{-1} = (A^{-1})^T$

## Types of Matrices

- a) Diagonal Matrices

$$D = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix}$$

$$D^{-1} = \begin{bmatrix} 1/d_1 & 0 & \cdots & 0 \\ 0 & 1/d_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1/d_n \end{bmatrix}$$

$$D^k = \begin{bmatrix} d_1^k & 0 & \cdots & 0 \\ 0 & d_2^k & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & d_n^k \end{bmatrix}$$

- b) Upper Triangular Matrix

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & a_{22} & a_{23} & a_{24} \\ 0 & 0 & a_{33} & a_{34} \\ 0 & 0 & 0 & a_{44} \end{bmatrix}$$

- c) Lower Triangular Matrix

$$\begin{bmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

## Transformations

- a)  $\mathbf{w} = A\mathbf{x} = T_A(\mathbf{x})$
- b)  $T_A(\mathbf{0}) = \mathbf{0}$
- c)  $T_A(k\mathbf{u}) = kT_A(\mathbf{u})$
- d)  $T_A(\mathbf{u} + \mathbf{v}) = T_A(\mathbf{u}) + T_A(\mathbf{v})$
- e)  $A = B$ , iff  $T_A(\mathbf{x}) = T_B(\mathbf{x})$
- f)  $A = [T(\mathbf{e}_1) \mid T(\mathbf{e}_2) \mid \dots \mid T(\mathbf{e}_n)]$
- g) Reflection Operators

| Operator                                                      | Illustration | Images of $\mathbf{e}_1$ and $\mathbf{e}_2$                                                                                             | Standard Matrix                                                      |
|---------------------------------------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Reflection about the $x$ -axis<br>$T(x, y) = (x, -y)$         |              | $T(\mathbf{e}_1) = T(1, 0) = (1, 0)$<br>$T(\mathbf{e}_2) = T(0, 1) = (0, -1)$                                                           | $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$                      |
| Reflection about the $y$ -axis<br>$T(x, y) = (-x, y)$         |              | $T(\mathbf{e}_1) = T(1, 0) = (-1, 0)$<br>$T(\mathbf{e}_2) = T(0, 1) = (0, 1)$                                                           | $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$                      |
| Reflection about the line $y = x$<br>$T(x, y) = (y, x)$       |              | $T(\mathbf{e}_1) = T(1, 0) = (0, 1)$<br>$T(\mathbf{e}_2) = T(0, 1) = (1, 0)$                                                            | $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$                       |
| Operator                                                      | Illustration | Images of $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$                                                                                    | Standard Matrix                                                      |
| Reflection about the $xy$ -plane<br>$T(x, y, z) = (x, y, -z)$ |              | $T(\mathbf{e}_1) = T(1, 0, 0) = (1, 0, 0)$<br>$T(\mathbf{e}_2) = T(0, 1, 0) = (0, 1, 0)$<br>$T(\mathbf{e}_3) = T(0, 0, 1) = (0, 0, -1)$ | $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ |
| Reflection about the $xz$ -plane<br>$T(x, y, z) = (x, -y, z)$ |              | $T(\mathbf{e}_1) = T(1, 0, 0) = (1, 0, 0)$<br>$T(\mathbf{e}_2) = T(0, 1, 0) = (0, -1, 0)$<br>$T(\mathbf{e}_3) = T(0, 0, 1) = (0, 0, 1)$ | $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ |
| Reflection about the $yz$ -plane<br>$T(x, y, z) = (-x, y, z)$ |              | $T(\mathbf{e}_1) = T(1, 0, 0) = (-1, 0, 0)$<br>$T(\mathbf{e}_2) = T(0, 1, 0) = (0, 1, 0)$<br>$T(\mathbf{e}_3) = T(0, 0, 1) = (0, 0, 1)$ | $\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ |

## h) Projection Operators

| Operator                                                                  | Illustration | Images of $\mathbf{e}_1$ and $\mathbf{e}_2$                                                                                            | Standard Matrix                                                     |
|---------------------------------------------------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Orthogonal projection<br>onto the $x$ -axis<br>$T(x, y) = (x, 0)$         |              | $T(\mathbf{e}_1) = T(1, 0) = (1, 0)$<br>$T(\mathbf{e}_2) = T(0, 1) = (0, 0)$                                                           | $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$                      |
| Orthogonal projection<br>onto the $y$ -axis<br>$T(x, y) = (0, y)$         |              | $T(\mathbf{e}_1) = T(1, 0) = (0, 0)$<br>$T(\mathbf{e}_2) = T(0, 1) = (0, 1)$                                                           | $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$                      |
| Operator                                                                  | Illustration | Images of $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$                                                                                   | Standard Matrix                                                     |
| Orthogonal projection<br>onto the $xy$ -plane<br>$T(x, y, z) = (x, y, 0)$ |              | $T(\mathbf{e}_1) = T(1, 0, 0) = (1, 0, 0)$<br>$T(\mathbf{e}_2) = T(0, 1, 0) = (0, 1, 0)$<br>$T(\mathbf{e}_3) = T(0, 0, 1) = (0, 0, 0)$ | $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ |
| Orthogonal projection<br>onto the $xz$ -plane<br>$T(x, y, z) = (x, 0, z)$ |              | $T(\mathbf{e}_1) = T(1, 0, 0) = (1, 0, 0)$<br>$T(\mathbf{e}_2) = T(0, 1, 0) = (0, 0, 0)$<br>$T(\mathbf{e}_3) = T(0, 0, 1) = (0, 0, 1)$ | $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ |
| Orthogonal projection<br>onto the $yz$ -plane<br>$T(x, y, z) = (0, y, z)$ |              | $T(\mathbf{e}_1) = T(1, 0, 0) = (0, 0, 0)$<br>$T(\mathbf{e}_2) = T(0, 1, 0) = (0, 1, 0)$<br>$T(\mathbf{e}_3) = T(0, 0, 1) = (0, 0, 1)$ | $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ |

## i) Rotation Operator

| Operator                                                                      | Illustration | Images of $\mathbf{e}_1$ and $\mathbf{e}_2$                                                                           | Standard Matrix                                                                         |
|-------------------------------------------------------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Counterclockwise<br>rotation about the<br>origin through an<br>angle $\theta$ |              | $T(\mathbf{e}_1) = T(1, 0) = (\cos \theta, \sin \theta)$<br>$T(\mathbf{e}_2) = T(0, 1) = (-\sin \theta, \cos \theta)$ | $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ |

## Chapter 2

- a)  $\det(A) = \det(A^T)$   
 b) Row reduction:

| Relationship                                                                                                                                                                                                                                                                        | Operation                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| $\begin{vmatrix} ka_{11} & ka_{12} & ka_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = k \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ $\det(B) = k \det(A)$                        | In the matrix $B$ the first row of $A$ was multiplied by $k$ .                    |
| $\begin{vmatrix} a_{21} & a_{22} & a_{23} \\ a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = - \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ $\det(B) = -\det(A)$                            | In the matrix $B$ the first and second rows of $A$ were interchanged.             |
| $\begin{vmatrix} a_{11} + ka_{21} & a_{12} + ka_{22} & a_{13} + ka_{23} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ $\det(B) = \det(A)$ | In the matrix $B$ a multiple of the second row of $A$ was added to the first row. |

- c)  $\det(kA) = k^n \det(A)$   
 d)  $\det(A + B) \neq \det(A) + \det(B)$   
 e)  $\det(AB) = \det(A) \det(B)$   
 f)  $\det(A^{-1}) = \frac{1}{\det(A)}$   
 g)  $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$

## Chapter 3

### Vector Properties

- a)  $\mathbf{P}_1\mathbf{P}_2 = \mathbf{OP}_2 - \mathbf{OP}_1 = (u_1, u_2, \dots, u_n) - (v_1, v_2, \dots, v_n)$   
 b)  $\mathbf{u} = k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \dots + k_r\mathbf{v}_r$   
 c)  $\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$   
 d)  $\|k\mathbf{v}\| = |k| \|\mathbf{v}\|$   
 e)  $\mathbf{u} = \frac{1}{\|\mathbf{v}\|} \mathbf{v}$   
 f)  $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + \dots + (u_n - v_n)^2}$   
 g)  $\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$   
 h)  $\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + \dots + u_nv_n$   
 i)  $\mathbf{v} \cdot \mathbf{v} = v_1^2 + v_2^2 + \dots + v_n^2 = \|\mathbf{v}\|^2$   
 j)  $\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}}$   
 k)  $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$   
 l)  $k(\mathbf{u} \cdot \mathbf{v}) = (k\mathbf{u}) \cdot \mathbf{v}$

- m)  $-1 \leq \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} \leq 1$
- n)  $|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|$  (Kashy Schwarzenigger)
- o)  $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$
- p)  $d(\mathbf{u}, \mathbf{v}) \leq d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v})$
- q)  $\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2)$
- r)  $|\mathbf{u} \cdot \mathbf{v}| = \frac{1}{4} \|\mathbf{u} + \mathbf{v}\|^2 - \frac{1}{4} \|\mathbf{u} - \mathbf{v}\|^2$
- s)  $\|\mathbf{u} + \mathbf{v}\|^2 = (\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v}) = \|\mathbf{u}\|^2 + 2(\mathbf{u} \cdot \mathbf{v}) + \|\mathbf{v}\|^2$
- t)  $A\mathbf{u} \cdot \mathbf{v} = \mathbf{u} \cdot A^T \mathbf{v}$
- u)  $\mathbf{u} \cdot A\mathbf{v} = A^T \mathbf{u} \cdot \mathbf{v}$
- v) Row vectors of A are  $\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_m$  and Column vectors of B are  $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n$

$$AB = \begin{bmatrix} \mathbf{r}_1 \cdot \mathbf{c}_1 & \mathbf{r}_1 \cdot \mathbf{c}_2 & \cdots & \mathbf{r}_1 \cdot \mathbf{c}_n \\ \mathbf{r}_2 \cdot \mathbf{c}_1 & \mathbf{r}_2 \cdot \mathbf{c}_2 & \cdots & \mathbf{r}_2 \cdot \mathbf{c}_n \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{r}_m \cdot \mathbf{c}_1 & \mathbf{r}_m \cdot \mathbf{c}_2 & \cdots & \mathbf{r}_m \cdot \mathbf{c}_n \end{bmatrix}$$

## Orthogonality

- a)  $\mathbf{u} \cdot \mathbf{v} = 0$
- b) Point Normal Equations where normal  $\mathbf{n} = (a, b)$  in  $R^2$  or  $(a, b, c)$  in  $R^3$
- 1)  $a(x - x_0) + b(y - y_0) = 0$  (line)
  - 2)  $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$  (plane)
- c)  $\mathbf{n} \cdot \mathbf{x} = 0$
- d) Projection formulas
- 1)  $\text{proj}_{\mathbf{a}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{a}}{\|\mathbf{a}\|^2} \mathbf{a}$  (component of  $\mathbf{u}$  along  $\mathbf{a}$ )
  - 2)  $\mathbf{u} - \text{proj}_{\mathbf{a}} \mathbf{u} = \mathbf{u} - \frac{\mathbf{u} \cdot \mathbf{a}}{\|\mathbf{a}\|^2} \mathbf{a}$  (component of  $\mathbf{u}$  orthogonal to  $\mathbf{a}$ )
- e)  $\|\text{proj}_{\mathbf{a}} \mathbf{u}\| = \frac{|\mathbf{u} \cdot \mathbf{a}|}{\|\mathbf{a}\|} = \|\mathbf{u}\| |\cos \theta|$
- f)  $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$
- g) Distance formulas
- 1)  $D = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$  (line)
  - 2)  $D = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}$  (plane)

## Geometry

- a)  $\mathbf{x} = \mathbf{x}_0 + t\mathbf{v}$  (line)
- b)  $\mathbf{x} = \mathbf{x}_0 + t_1\mathbf{v}_1 + t_2\mathbf{v}_2$  (plane)
- c)  $\mathbf{x} = \mathbf{x}_0 + t_1\mathbf{v}_1 + \dots + t_n\mathbf{v}_n$  (hyperplane of dim n)
- d)  $\mathbf{x} = \mathbf{x}_0 + t(\mathbf{x}_1 - \mathbf{x}_0) = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1$  (two-point line equation)
- e) Dot Product form of  $A\mathbf{x} = \mathbf{0}$
- $$\begin{aligned} \mathbf{r}_1 \cdot \mathbf{x} &= 0 \\ \mathbf{r}_2 \cdot \mathbf{x} &= 0 \\ \vdots & \\ \mathbf{r}_m \cdot \mathbf{x} &= 0 \end{aligned}$$

## Cross Product

- a)  $\mathbf{u} \times \mathbf{v} = (u_2v_3 - u_3v_2, u_3v_1 - u_1v_3, u_1v_2 - u_2v_1) = \left( \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix}, -\begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix}, \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \right)$
- b)  $\mathbf{u} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{u})$
- c)  $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = (\mathbf{u} \times \mathbf{v}) + (\mathbf{u} \times \mathbf{w})$
- d)  $(\mathbf{u} + \mathbf{v}) \times \mathbf{w} = (\mathbf{u} \times \mathbf{w}) + (\mathbf{v} \times \mathbf{w})$
- e)  $k(\mathbf{u} \times \mathbf{v}) = (k\mathbf{u}) \times \mathbf{v} = \mathbf{u} \times (k\mathbf{v})$
- f)  $\mathbf{u} \times \mathbf{0} = \mathbf{0} \times \mathbf{u} = \mathbf{0}$
- g)  $\mathbf{u} \times \mathbf{u} = \mathbf{0}$
- h)  $\mathbf{u} \cdot (\mathbf{u} \times \mathbf{v}) = \mathbf{0}$
- i)  $\mathbf{v} \cdot (\mathbf{u} \times \mathbf{v}) = \mathbf{0}$
- j)  $\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\|^2 \|\mathbf{v}\|^2 - (\mathbf{u} \cdot \mathbf{v})^2$  (Lagrange's Identity)
- k)  $\mathbf{u} \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{u} \cdot \mathbf{v})\mathbf{w}$
- l)  $(\mathbf{u} \times \mathbf{v}) \times \mathbf{w} = (\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{v} \cdot \mathbf{w})\mathbf{u}$
- m)  $\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \mathbf{k}$
- n)  $\mathbf{i} \times \mathbf{j} = \mathbf{k}, \mathbf{j} \times \mathbf{k} = \mathbf{i}, \mathbf{k} \times \mathbf{i} = \mathbf{j}$
- o)  $\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| \sin \theta$  (also Area of Parallelogram)
- p) Area of Triangle =  $\frac{1}{2} \|\mathbf{u} \times \mathbf{v}\|$
- q)  $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} = \mathbf{w} \cdot (\mathbf{u} \times \mathbf{v}) = \mathbf{v} \cdot (\mathbf{w} \times \mathbf{u})$  (also Volume of Parallelepiped)
- r) Volume of Trapezoid =  $\frac{1}{6} \|\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})\|$

## Chapter 4

### Vector Space Axioms

1. If  $\mathbf{u}$  and  $\mathbf{v}$  are objects in  $V$ , then  $\mathbf{u} + \mathbf{v}$  is in  $V$ .
2.  $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
3.  $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
4. There exists an object in  $V$ , called the **zero vector**, that is denoted by  $\mathbf{0}$  and has the property that  $\mathbf{0} + \mathbf{u} = \mathbf{u} + \mathbf{0} = \mathbf{u}$  for all  $\mathbf{u}$  in  $V$ .
5. For each  $\mathbf{u}$  in  $V$ , there is an object  $-\mathbf{u}$  in  $V$ , called a **negative** of  $\mathbf{u}$ , such that  $\mathbf{u} + (-\mathbf{u}) = (-\mathbf{u}) + \mathbf{u} = \mathbf{0}$ .
6. If  $k$  is any scalar and  $\mathbf{u}$  is any object in  $V$ , then  $k\mathbf{u}$  is in  $V$ .
7.  $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$
8.  $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$
9.  $k(m\mathbf{u}) = (km)\mathbf{u}$
10.  $1\mathbf{u} = \mathbf{u}$

## Subspace Test

- a) If  $\mathbf{u}$  and  $\mathbf{v}$  are vectors in  $W$ , then  $\mathbf{u} + \mathbf{v}$  is in  $W$ .
- b) If  $k$  is a scalar and  $\mathbf{u}$  is a vector in  $W$ , then  $k\mathbf{u}$  is in  $W$ .
- c) Existence of **zero vector** in  $W$ . (Extra)

## Basis

- a)  $\mathbf{b} = (b_1, b_2, \dots, b_n) = k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \dots + k_n\mathbf{v}_n$  (Spanning Check)
- b)  $\mathbf{0} = k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \dots + k_n\mathbf{v}_n$ , iff  $(k_1, \dots, k_n) = (0, \dots, 0)$  (Linear Independence Check)

## Wronskian

$$W(x) = \begin{vmatrix} f_1(x) & f_2(x) & \cdots & f_n(x) \\ f_1'(x) & f_2'(x) & \cdots & f_n'(x) \\ \vdots & \vdots & & \vdots \\ f_1^{(n-1)}(x) & f_2^{(n-1)}(x) & \cdots & f_n^{(n-1)}(x) \end{vmatrix}$$

$\det(W(x)) \neq 0 \Rightarrow$  Linear Independent

## Dimension & Spaces

- a) Different Dimensions
  - 1)  $\dim(R^n) = n$
  - 2)  $\dim(P_n) = n + 1$
  - 3)  $\dim(M_{mn}) = mn$
- b) Space Properties
  - 1)  $A\mathbf{x} = \mathbf{b}$  is consistent iff  $\mathbf{b}$  is in column space of  $A$ .
  - 2) Row equivalent matrices have the same row space.
  - 3) Row equivalent matrices have the same null space.
  - 4) Row and Column space of a matrix have the same dimension.
  - 5)  $\text{rank}(A) + \text{nullity}(A) = n$
  - 6)  $\text{rank}(A^T) + \text{nullity}(A^T) = m$